

Search APIs Are Decision Surfaces for Tool-Using Language Agents

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Abstract

Search APIs are the fundamental retrieval layer and often the most often used tool by agents. Traditional Search APIs provide URL, title and snippet allowing a preview of contents from the website. Owing to token cost constraints, incremental disclosure by allowing the agent to fetch full page only after introduction of a snippet is slowly becoming the norm in agent retrieval architecture. In such systems, when evaluating performance of Search APIs, answer accuracy is often considered as primary metric. We argue that a commercial search API is better understood as a *decision surface*: the ranked snippets, URLs, and metadata that determine whether an agent answers immediately, searches again, or spends tokens opening pages. We test this claim with one frozen GPT-5.4 agent, two tools (search_web and fetch_page), and 100 questions from SEALQA-HARD, varying only the search provider (Brave, Tavily, Firecrawl). A Kimi-K2.6 oracle labels every element (URL, title, snippet and fetched page (if fetched)) content visible to the agent, producing 6,869 valid per-URL judgments. We use an audited *correct-answer* label, *semantic_match*, which preserves exact matches while accepting harmless formatting and naming variants. Under this measure the providers remain close (25, 25, 26 / 100), but their evidence economies differ sharply: by offering gold information rich snippets (Brave), concentrated gold supporting URLs at rank 1 (Tavily) or encouraging broad exploration (Firecrawl). We also introduce an agent answer agnostic metric: contradict-to-gold URL ratio, that varies from 0.92 to 2.59. Provider choice is therefore a retrieval-budget and policy decision, not merely a recall decision.

1 Introduction

Early versions of search-augmented agents were primarily dependent on the extracted page data that were provided by Search API providers. As complex use cases have evolved over time, the agent

landscape is shifting toward progressive disclosure approach where the models are provided top-n URLs, title, snippet and key metadata like freshness, etc. In this new landscape, the engineering practices often treats Search API providers as a replaceable component. If two APIs return plausible top- k URLs and produce similar answer accuracy, the rest of the agent pipeline is assumed to be largely unaffected.

This paper argues that the relevant object is the *pre-fetch surface*. Before a page is opened, an agent does not see a search index or a corpus and relies on Search API results. That surface is the evidence state on which the agent decides whether to answer, search again, or spend fetch tokens. We call commercial search APIs **decision surfaces** for tool-using language agents.

The distinction matters because final correctness can converge while the internal pipeline diverges. A provider may expose enough snippet evidence for immediate answering. Another may put the relevant URL at rank 1, making a top-result fetch policy effective. A third may expose sparse snippets but encourage broad exploration. These regimes can yield similar aggregate accuracy with different cost, latency, contamination, and failure modes.

We test this claim under a controlled protocol, we freeze the answer model, prompt, tools, maximum iterations, judge, and page-fetch backend. The only experimental condition is the commercial search API: Brave, Tavily, or Firecrawl. We then judge every data visible to the agent. The per-URL oracle lets us ask not only whether the final answer was correct, but what evidence and actions were available at the decision surface.

Contributions.

1. We formalize commercial search APIs as *decision surfaces* for tool-using agents rather than static ranked-list retrievers.
2. We introduce a per-URL oracle protocol that

labels every API result element the agent saw, including fetched pages.

3. We define surface-visible support, a four-way decision partition (SMART, MISSED, BLIND, NO-OP), and a contradiction-to-gold metric that is independent of the model answer.
4. We show that similar correctness hides different provider-induced retrieval regimes: visible support, rank concentration, blind exploration, contamination, and complementarity.

2 Related Work

Retrieval-augmented generation and open-domain QA. Retrieval-augmented generation combines parametric models with non-parametric evidence (Lewis et al., 2020; Guu et al., 2020). Open-domain QA systems such as DPR, FiD, and Atlas largely evaluate whether a retriever supplies answer-bearing passages to a reader or generator (Karpukhin et al., 2020; Izacard and Grave, 2021; Izacard et al., 2023). Our setting differs in two ways: retrieval is performed by production web-search APIs, and the reader is an agent that can choose whether to fetch pages.

Information-retrieval evaluation. Classical IR metrics, including precision/recall and rank-aware measures such as NDCG, evaluate static rankings (Järvelin and Kekkonen, 2002; Manning et al., 2008). BEIR broadened zero-shot retriever evaluation across tasks and domains (Thakur et al., 2021). Such metrics are necessary but incomplete for agents because they do not ask whether the pre-fetch surface was sufficient, misleading, or action-guiding.

Tool-using and browsing agents. WebGPT, Self-Ask, ReAct, IRCOT, Toolformer, and Self-RAG study language models that search, browse, or decide when to call tools (Nakano et al., 2021; Press et al., 2022; Yao et al., 2023; Trivedi et al., 2022; Schick et al., 2023; Asai et al., 2024). FreshLLMs shows that search augmentation helps with fresh factual knowledge and that evidence order matters (Vu et al., 2023). Over-searching work argues for cost-sensitive evaluation of search-augmented models (Xie et al., 2026). We isolate a complementary variable: the provider surface presented to the same agent.

Hard search benchmarks and evidence evaluation. GAIA, WebArena, BrowseComp, and

SEALQA evaluate agents or systems on tasks requiring search, browsing, or hard factual reasoning (Mialon et al., 2024; Zhou et al., 2024; Wei et al., 2025; Pham et al., 2025). RAG evaluation frameworks such as RAGAS and ARES evaluate answer quality, faithfulness, context relevance, or generated claims (Es et al., 2024; Saad-Falcon et al., 2024). We use a hard-QA benchmark as a controlled probe for provider-induced evidence states.

Attribution, verifiability, and LLM judges.

Generative search systems may produce fluent answers with incomplete citation support (Liu et al., 2023a; Yue et al., 2023); long-context work shows that more text does not guarantee robust evidence use (Liu et al., 2024). LLM-as-judge methods scale evaluation but require constrained rubrics and careful interpretation (Zheng et al., 2023; Liu et al., 2023b). Our judge labels individual retrieved documents with a fixed JSON schema, and our answer correctness label is separately audited as `semantic_match`.

Commercial search APIs. Brave, Tavily, Firecrawl, and Jina Reader expose different product surfaces (Brave Software, 2026; Tavily, 2026; Firecrawl, 2026; Jina AI, 2026). We disable provider-side page content in the main condition and use a shared `fetch_page` backend, so provider differences are attributable to ranked URLs, snippets, and metadata rather than distinct extraction pipelines.

3 Experimental Protocol

Figure 1 summarizes the experiment.

Dataset. We use a deterministic proportional stratified sample of 100 questions from the 254-row SEALQA-HARD subset. The strata are `freshness × search_results × topic`, with seed 20260509. Appendix B gives the full sampling summary.

Agent and tools. The answer model is GPT-5.4 with a maximum of 10 iterations. It has two tools. `search_web` accepts a query and returns up to ten ranked results, each with a document identifier, rank, title, URL, domain, snippet surface, and provider metadata. `fetch_page` accepts a document identifier from a prior search result and returns extracted markdown plus fetch metadata. The tool interface allows the agent to issue *multiple* `search_web` calls or *multiple* `fetch_page` calls in a single turn; the trace records each call separately.

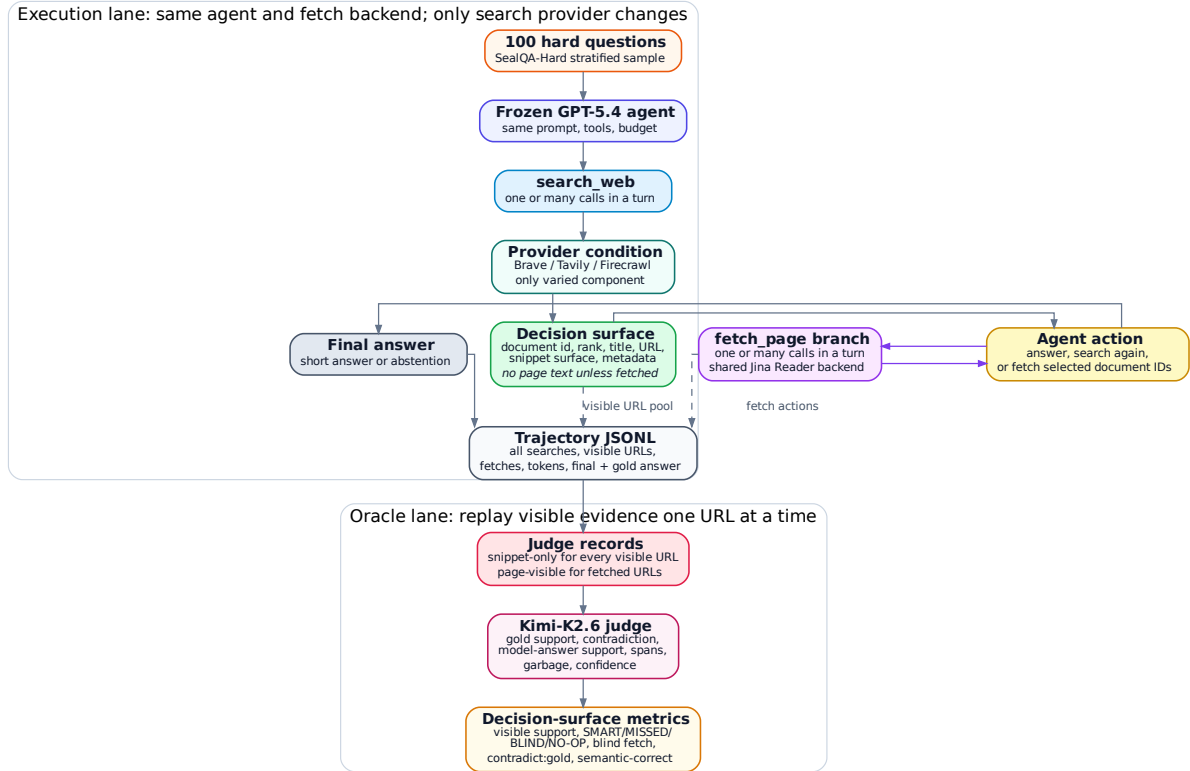


Figure 1: Execution and judging pipeline. The provider returns only the pre-fetch decision surface: ranked URLs, snippets, and metadata. The agent may issue multiple `search_web` or `fetch_page` calls in one turn. Every visible URL is replayed to the Kimi judge in the same representation the answer model saw, yielding per-URL oracle labels.

The final answer must use the exact prefix `FINAL ANSWER:` and be grounded in retrieved evidence.

Providers and snippet normalization. We compare Brave, Tavily, and Firecrawl Search. Each adapter normalizes provider output into the same result schema. Brave exposes provider-native additional snippet blocks. We aggregate those into the generic *snippet surface* rather than treating them as a separate evidence channel; the agent saw them as snippets, and the judge saw the same text. We still discuss this asymmetry as a limitation because it changes the amount of pre-fetch text Brave returns. Tavily is run with raw content disabled, and Firecrawl is run without provider-side markdown scraping. Page text visible to the agent therefore comes only from the shared Jina Reader backend.

Traces. The experiment produces 300 provider-query trajectories. A trace records every search call, every returned URL, every fetch decision, fetch status, token usage, latency, final answer, and gold answer. Provider-comparison scripts deterministically derive token, URL-hit, and per-query support fields from these traces. Answer correctness comes from the separate semantic audit TSV.

4 Per-URL Oracle and Metrics

Judge input. For every URL the agent saw, we run Kimi-K2.6 from Azure as the judge. The judge receives the question, gold answer, model final answer, and one retrieved document in the same XML-like format the answer model saw. For snippet-only records, the judge sees title, URL, domain, snippet surface, and metadata. For page-visible records, it also sees extracted markdown.

Judge output. The judge returns JSON only and is instructed not to use outside knowledge. The required schema is shown below.

Per-URL judge output JSON.

- `contains_gold_answer,` `gold_answer_in_snippets,`
- `gold_answer_in_extracted_page`
- `supports_model_answer,` `contradicts_gold_answer,`
- `is_garbage`
- `garbage_reason,` `answer_span,` `gold_snippet_span,`
- `gold_extracted_page_span,` `confidence`

Across providers, 6,869 of 6,909 judge rows are valid; 40 invalid rows are excluded. Valid rows split into 6,519 snippet-only and 350 page-visible rows.

Visible support. For a provider-query pair, visible support exists if at least one valid judge row has

Metric	Brave	Tavily	Firecrawl
CORRECT /100	25	25	26
Exact match /100	21	21	23
Semantic gain	+4	+4	+3
F1	0.270	0.261	0.282
Avg. search calls	2.29	2.74	2.51
Avg. fetch calls	1.02	1.30	1.28
Fetches queries	65%	76%	81%
Tokens/query	59,627	54,156	57,979
URL hit	59	57	60
Domain hit	82	82	82
Snippet-surface string hit	71	60	54
Page string hit	52	57	61

Table 1: Headline metrics. CORRECT is the audited semantic match; EM is the stricter normalized string metric. The snippet-surface row aggregates all provider-returned snippet text, including Brave’s native additional snippets.

contains_gold_answer true on either the snippet-only surface or a page-visible surface. This is a lower bound on true provider pool support because unfetched pages are not page-judged.

Decision partition. We join the per-URL oracle with trace actions and assign each provider-query pair to one cell: SMART if visible support exists and the agent fetched a gold-supporting URL; MISSED if visible support exists and the agent fetched none of the supporting URLs; BLIND if no visible support exists and the agent fetched at least one URL; and NO-OP if no visible support exists and no URL was fetched. The partition is descriptive, not causal: a MISSED query may still be answered correctly from snippets.

Answer-independent contamination. We define the contradict-to-gold ratio over snippet-only URL rows:

$$r_{c:g} = \frac{|\{u : \text{contradicts_gold_answer}(u)\}|}{|\{u : \text{contains_gold_answer}(u)\}|}.$$

It uses only surface labels, not the model’s answer or page text.

5 Results

5.1 Correctness parity masks different evidence economies

Table 1 and Figure 2 show the headline. Under CORRECT, the providers remain close: 25, 25, and 26 correct answers out of 100. The semantic audit changes absolute counts, but not the main conclusion: aggregate correctness alone suggests the providers are nearly interchangeable.

Metric	Brave	Tavily	Firecrawl
Visible support	33	24	19
No visible support	67	76	81
Snippet rows	2,095	2,339	2,085
Page-visible rows	101	125	124
Rank-1 gold rows	12 (12%)	15 (48%)	3 (11%)
$r_{c:g}$	0.92	1.87	2.59

Table 2: Visible support and surface diagnostics from the per-URL judge. Visible support is a lower bound because only fetched URLs receive page-visible judgments.

The evidence economy says otherwise. Brave exposes visible support on 33 queries, compared with 24 for Tavily and 19 for Firecrawl. Tavily has much stronger rank concentration: 15 gold-supporting snippet rows at rank 1 (48%), versus 12% and 11% for the others. Firecrawl fetches on the largest share of queries (81%), consistent with a volume/exploration regime.

5.2 The visible-support ceiling moves sharply

Table 2 shows that the per-URL oracle sees a larger provider effect than headline correctness. The surface-visible support counts differ by 14 queries between Brave and Firecrawl, while final correct answers differ by one. The contradiction ratio also varies substantially, from 0.92 to 2.59 contradicting rows per gold-supporting row.

5.3 The same agent enters different decision regimes

Figure 3 shows the mechanism. Brave’s support-visible mass is larger, but many such queries are MISSED rather than SMART, consistent with a snippet-rich surface where fetching is not always needed. Tavily’s support is smaller but rank concentrated, making SMART fetches higher-yield. Firecrawl’s largest regime is BLIND exploration: no support was visible to the judge before fetch, yet the agent frequently opened pages.

effig:partition; Wilson intervals are in Appendix efapp:wilson.

5.4 Aggregate correctness hides complementarity

Only 10 questions are answered correctly by all three providers. 12 are correct under exactly two providers, 22 under exactly one, and 56 under none. Thus provider choice changes *which* questions are solved, not only how many. This suggests that provider routing or provider-aware fetch policies may matter more than choosing a single winner.

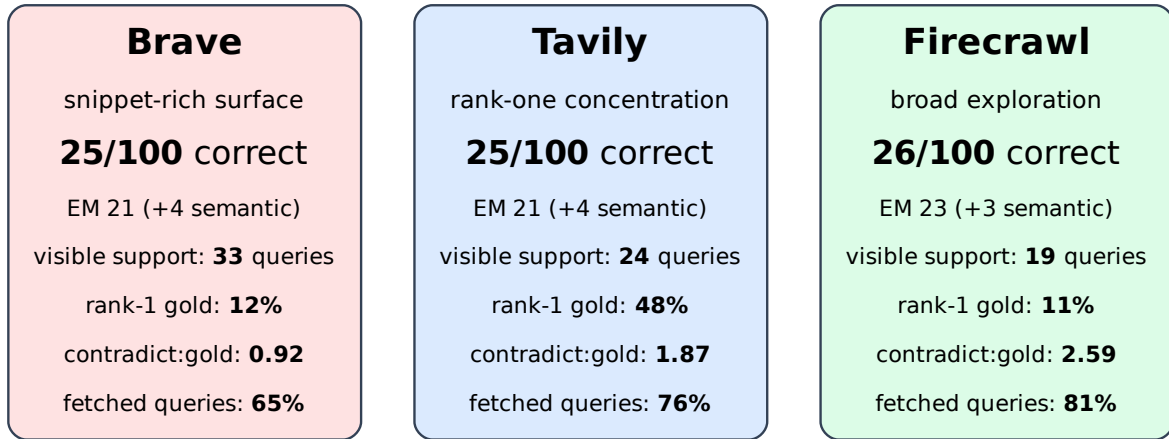


Figure 2: Provider profiles under the semantic CORRECT metric. EM is shown only as an audit column. Similar correctness hides different visible support, rank concentration, contamination, and fetch behavior.

Provider	SMART support visible + fetched support	MISSED support visible + did not fetch it	BLIND no visible support + fetched	NO-OP no visible support + no fetch
Brave	8 / 3 (38%)	25 / 9 (36%)	51 / 9 (18%)	16 / 0 (0%)
Tavily	11 / 7 (64%)	13 / 5 (38%)	55 / 9 (16%)	21 / 0 (0%)
Firecrawl	7 / 2 (29%)	12 / 5 (42%)	67 / 16 (24%)	14 / 0 (0%)

Figure 3: Four-way decision partition. Each entry is queries / correct answers (semantic match) after joining oracle labels with agent fetch actions.

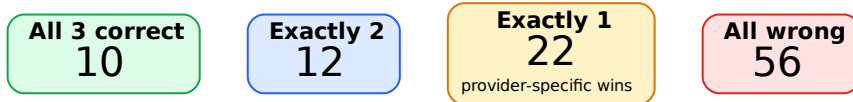


Figure 4: Semantic-correctness overlap across providers. An oracle provider router would solve 44 queries, above the best single provider.

5.5 Many failures occur despite visible answer text

The provider-comparison traces include a deterministic proxy for whether the gold answer string appears anywhere in retrieved text. This is *not* the Kimi judge and should not be confused with document support. It is a normalized substring check over title, snippets, provider-returned snippet text, and fetched page text, with a punctuation-sensitive fallback for answers such as numbers. Under this string proxy, answer text is visible somewhere for 78, 75, and 76 provider-query pairs. Yet the agent still has 57, 56, and 53 wrong answers despite that text being present.

This failure mode reinforces the core claim. A search surface does not merely decide whether evidence exists; it shapes whether the model notices, trusts, fetches, and uses the right evidence amid distractors.

6 Discussion

Provider choice is a policy choice. The same agent policy has different utility under different provider surfaces. Tavily’s rank-one concentration rewards top-result fetching. Brave’s richer snippet surface reduces the marginal value of fetching some support-bearing pages. Firecrawl induces broader exploration. A provider should therefore be paired with a provider-aware fetch policy, not a fixed universal heuristic.

Top- k relevance is incomplete for agents. Static relevance and gold URL hit rates remain useful, but they cannot see whether the surface was already answerable, whether a fetch was worthwhile, or whether contradictory snippets competed with gold support. Decision-surface metrics complement IR metrics by measuring the action state available before the fetch decision.

What providers could optimize. An agent-ready search API should optimize not only relevance but also actionability: calibrated snippets, stable document identifiers, source/freshness meta-data, low contradiction-to-gold contamination, and rankings that align answer-bearing pages with common agent fetch policies.

What practitioners can use now. The metrics here can be computed from traces without rerunning provider APIs. A practitioner can run a small sample through candidate providers, judge visible URLs, and choose a policy based on whether the provider behaves like a snippet-rich surface, a rank-concentrated surface, or a volume/exploration surface.

7 Limitations

Single agent and judge. The decision partition is induced by one GPT-5.4 agent and one Kimi-K2.6 judge. A different model, prompt, or fetch appetite may shift the cell counts.

Lower-bound support. Visible support is a lower bound: unfetched URLs are not page-judged. This prevents us from claiming full provider recall.

Snippet-surface asymmetry. Brave returns more provider-native snippet text under our configuration. We aggregate it into the snippet surface to avoid presenting it as special treatment, but it remains a real product-surface difference and a plausible driver of Brave’s visible-support ceiling.

Observational policy analysis. The agent’s actions are observed, not randomized. We do not intervene on ranks, snippets, or fetch decisions. Causal replay is future work.

Scale and provider coverage. The main experiment has 100 questions and three providers. Small cells should be read as diagnostic rather than definitive. Additional providers such as Bing, Google CSE, Serper, Exa, and others would broaden the claim.

8 Conclusion

Commercial search APIs are not interchangeable input pipes for tool-using language agents. In this controlled study, semantic-correct answer rates are close, but the internal retrieval economy changes

sharply: visible support, rank concentration, contradiction, fetch behavior, and instance-level correctness all depend on the provider surface. Search APIs for agents should therefore be evaluated as decision surfaces: ranked, snippet-bearing action states that allocate retrieval budget.

Ethics Statement

This work evaluates commercial search APIs and LLM agents on public web-search QA data. We do not claim that one provider is universally better. Provider names are reported because they are experimental conditions and because reproduction requires knowing which APIs produced the surfaces. The study may help reduce unnecessary page fetches and improve handling of contradictory evidence. It could also be misread as a provider leaderboard without respecting the stated limitations; we discourage that use.

Reproducibility Statement

All paper numbers are deterministic functions of the committed query sample, trace JSONLs, judge JSONLs, provider-comparison summaries, and semantic audit TSV. The raw trace and judge files are stored through Git LFS; pull LFS artifacts before running the figure script. Final make targets fail if raw judge files are still LFS pointers, preventing accidental submission from incomplete data.

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A Formal Metric Definitions

For a provider p and query q , let $U_{p,q}$ denote the valid judged URLs visible in the trajectory. Let $G(u)$ be the event that the judge marks `contains_gold_answer` true on either the snippet-only record or a page-visible record, and let $F(u)$ indicate that the agent fetched URL u . A query has *visible support* when some $u \in U_{p,q}$ satisfies $G(u)$. It is **SMART** when visible support exists and a gold-supporting URL was fetched; **MISSED** when visible support exists but no gold-supporting URL was fetched; **BLIND** when no visible support exists but the agent fetched at least one URL; and **NO-OP** when no visible support exists and the agent fetched nothing. The contradiction ratio is computed over snippet-only rows as contradicting URLs divided by gold-supporting URLs.

B Dataset Stratification

The sample file records dataset `vtllms/sealqa`, subset `seal_hard`, sample size 100, source size 254, seed 20260509, and strata `freshness`, `search_results`, and `topic`. Marginal counts are: 25 fast-changing, 43 slow-changing, 32 never-changing; 57 conflicting and 43 unhelpful search-result labels; topics are Science & Technology 27, Sports 22, Entertainment 22, Others 12, Politics 9, and History & Geography 8. Effective years are 61 before 2024, 16 in 2024, and 23 in 2025. Multi-label question-type tags include 72 advanced reasoning, 61 entity/event disambiguation, 13 temporal tracking, 5 cross-lingual, and 2 false-premise.

C Provider Request Configurations

All providers are configured to return web-search results rather than provider-side page extracts. Brave uses its Web Search GET endpoint with count 10, offset 0, US/English settings, moderate safe search, spellcheck enabled, text decorations disabled, and web-only result filtering. Tavily uses its search endpoint with `search_depth=basic`, `topic=general`, `max_results=10`, and provider-side answer, image, favicon, and raw-content fields disabled. Firecrawl uses its v2 search endpoint with limit 10, web source only, US country, 60s timeout, and no provider-side markdown scraping. Page text for all provider conditions comes only from the shared Jina Reader backend called through `fetch_page`.

D Judge Prompt and Output Schema

The judge prompt instructs Kimi-K2.6 to use only the supplied document content and to return JSON only. A valid row has the expected schema version, a parsed judgment dictionary, provider/query/retrieval/URL identifiers, and no execution or parse error. Invalid rows are excluded rather than counted as negative evidence. The output schema includes gold-support fields, model-answer support, contradiction, garbage, short evidence spans, and confidence; Section 4 lists the exact JSON field names.

E Example Judgments and Semantic Corrections

The semantic audit marks some EM misses as correct when the answer is semantically equivalent. Representative examples are: Brave *Astra Zeneca* vs. *AstraZeneca*; Brave *UnionPay* vs. *China UnionPay*; Tavily *3 players* vs. *3*; Firecrawl *Bohemian Rhapsody* vs. *Bohemian Rhapsody, 9,948,386 viewers*; and Firecrawl *16 years* vs. *16 years old*. The regenerated audit file records the exact TSV rows and judge notes. Per-URL judge examples can be inspected by joining the judge JSONLs with the trace viewer: support-bearing rows have `contains_gold_answer` true and a span; contamination rows have `contradicts_gold_answer` true; relevant-but-insufficient rows remain false for both support fields.

F Trace Viewer and Qualitative Audit

The repository includes human-readable trajectory views in `results/trace_views/`. The rendering script is `scripts/render_trace.py`, and `scripts/trace_dashboard.py` provides a local browsing dashboard. These views were used for spot-checking case studies, verifying multi-call turns, and inspecting failures where answer text was present but unused. They are not an independent metric source: the paper numbers come from trace JSONLs, judge JSONLs, the semantic TSV, and provider-comparison summaries.

G Additional Diagnostics and Artifact Manifest

The tables below report Wilson intervals for the main decision cells, pairwise provider complementarity, and token/fetch diagnostics.

Provider/cell	Correct rate and Wilson 95% CI
Brave SMART	3/8: 38% [14–69]
Brave MISSED	9/25: 36% [20–55]
Brave BLIND	9/51: 18% [10–30]
Tavily SMART	7/11: 64% [35–85]
Tavily MISSED	5/13: 38% [18–64]
Tavily BLIND	9/55: 16% [9–28]
Firecrawl SMART	2/7: 29% [8–64]
Firecrawl MISSED	5/12: 42% [19–68]
Firecrawl BLIND	16/67: 24% [15–35]

Table 3: Wilson 95% intervals for decision-cell correctness. NO-OP rates are 0% for all providers in this run.

Pair	both	left only	right only	both wrong
Brave vs. Firecrawl	14	11	12	63
Brave vs. Tavily	14	11	11	64
Firecrawl vs. Tavily	14	12	11	63

Table 4: Pairwise semantic-correct overlap derived from the semantic audit TSV.

Metric	Brave	Tavily	Firecrawl
Total tokens	5.96M	5.42M	5.80M
Median/query	40,638	36,867	34,383
Max/query	303,462	305,474	380,646
>100k queries	19	16	16
Fetch success/fail	92/10	119/11	121/7

Table 5: Token and fetch diagnostics from provider summaries.

Raw artifacts. Audit mode consumes the semantic audit TSV, three phase-1 trace JSONLs, three Kimi per-URL judge JSONLs, and the provider-comparison summary artifacts. The script checks for Git LFS pointer files and fails explicitly if raw data has not been pulled.